# Project Report

**On**

# ArogyaX-AI-Medical-Chatbot

**Session: JAN 2026 – May 2026**

**Supervised by:**

DR Prashant Shukla

**Submitted by:**


1. **Ayush Upadhyay E23CSEU1146**
2. **Riti Dubey E23CSEU1146**
3. **Gaurav E23CSEU1152**
4. **Tanistha Keshri E23CSEU1165**


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Bachelor Of Technology CSE (AIML)


**SCHOOL OF COMPUTER SCIENCE ENGINEERING AND TECHNOLOGY,**

**BENNETT UNIVERSITY**

# CERTIFICATE

We hereby declare that the research project titled **"ArogyaX-AI-Medical-Chatbot"** is an original work carried out by us under the guidance of Dr. Prashant Shukla.

This progress report is a bona fide record of the work completed by us during the period from January 2026 to April 2026. The work presented in this report has not been submitted earlier to any institution or university for the award of any degree or diploma.

We further declare that all sources of information used in this report have been properly acknowledged and cited.

**Signatures of Students**

5. Ayush Upadhyay E23CSEU1146
6. Riti Dubey E23CSEU1146
7. Gaurav E23CSEU1152
8. Tanistha Keshri E23CSEU1165

# ACKNOWLEDGEMENT

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Additionally, we would like to acknowledge our institution for providing us with the platform and facilities to successfully complete this project.

## Abstract

The rapidly evolving technologies, especially those concerning AI, have had a major impact on the healthcare sector by enabling intelligent machines to make the task convenient for users. In the case of the suggested project, the development of an AI medical chatbot called ArogyaX utilizes advanced AI technologies in order to develop a medical assistant system.

For instance, NLP will be applied alongside retrieval-augmented generation (RAG) that will enable the system to understand the query of the user and respond appropriately based on its context. Moreover, OCR and image processing will be applied in order to enable the system to receive different forms of inputs, such as textual queries, PDF files containing the results of a medical report, written prescriptions, and even images regarding the issue the user is asking about.

It will make the system more reliable since it will be able to access related information before providing an answer. Various inputs were received in order to test the significance of the system.

This will increase the reliability of the system since it will be able to look for relevant information prior to answering the question. Various user inputs were tried on the developed system that helped to prove the significance of the system.

# Chapter 1: Introduction to the Project

Introduction

With the rapid progress of Artificial Intelligence (AI), different industries have witnessed substantial changes due to the adoption of AI technology. One of the areas that have significantly benefited from the development of Artificial Intelligence is the field of healthcare. Recently, AI-powered platforms like medical chatbots have been developed to provide timely health assistance using NLP (Natural Language Processing), ML (Machine Learning) and Generative AI technologies.

The main purpose of developing the project called ArogyaX – AI Medical Chatbot is to create an intelligent platform for the provision of medical assistance. This project involves the creation of a system through which the user would be able to interact with the application, receiving helpful responses from it, based on his or her medical query. The key idea behind the development of this application lies in connecting the user and his or her medical problems to receive some assistance in interpreting their situation.

It is important to mention that traditional healthcare services are limited and require more efforts than they could be provided by AI. First of all, it refers to the time needed to consult a doctor or another specialist in order to get medical help.

The ArogyaX platform incorporates cutting-edge AI methodologies to improve user engagement and offer more precise and relevant answers. The platform uses a chatbot that supports not only text input but also other forms of media, such as reports and images. This feature makes the chatbot a more flexible tool than rule-based solutions that only support text inputs.

In summary, the project shows how AI can be applied for the development of a smart and effective healthcare assistance system. Despite not being able to substitute the work of professionals, the platform offers an essential preliminary stage in providing help and raising awareness about diseases.

## 1.1 Background of the Study

With the introduction of AI into the healthcare sector, there have been several possibilities for the improvement of efficiency and effectiveness in the process. With increasing demands for immediate healthcare services, the conventional method encounters various restrictions such as the availability of healthcare centers, delays, and unavailability of consultation at distant places.

There has been an increasing application of intelligent chatbots with the capacity to interact in the form of natural language. Such intelligent chatbots utilize technologies like NLP, Machine Learning, and Deep Learning to understand the query of the user and give

appropriate responses. Intelligent chatbots can be applied in the medical sector to assist users in their health analysis, answering their queries, and analyzing their basic health reports.

“ArogyaX – AI Medical Chatbot” is a proposed project to develop an AI-based medical chatbot to provide medical services to users immediately.

## 1.2 Problem Statement

Although many developments have occurred in the field of medicine, the following problems still persist:

The availability of medical personnel becomes challenging in far-off places or underdeveloped communities.

Losing time while waiting for a consultation.

Struggling to understand the report and prescription prepared by the doctor.

Instant assistance not available for minor diseases.

Improper response to mass inquiries raised by patients.

Current systems either lack specificity, rely on rules-based techniques, or are incapable of understanding complicated medical questions. Additionally, most systems cannot interpret medical documents such as reports and handwritten prescriptions.

A need exists for an intelligent and efficient system that can:

- Analyze medical questions.
- Respond to medical questions.
- Provide interpretation of medical documents.

## 1.3 Objectives of the Project

Among the chief objectives of the ArogyaX - AI Medical Chatbot project are the following:

- Creation of a healthcare chatbot based on AI technology that will have capability of answering medical queries raised by patients.
- Ensuring immediate availability of correct health information to the users.
- Assistance of the users in analysing the symptoms and giving some suggestions for diagnoses.
- Processing of health reports which may be in a written format too.
- Processing of images such as X-rays.
- User interface creation.

### 1.4 Methodology Overview

ArogyaX – AI Medical Chatbot uses a modular and input-oriented methodology where the input generated by the user in real-time is taken into account in order to understand and act accordingly. Input types can vary widely; some of which may include text inputs, medical documents such as PDFs and hand-written prescriptions, and images including X-rays.

From the point of view of overall methodology used, it begins from taking in user input, followed by data extraction through NLP and OCR, then preprocessing of data, converting data into vectors for obtaining semantics and finally retrieving medical data through a retrieval mechanism wherein AI models provide contextually aware outputs.

The project is built using high-end technologies including Python programming language, transformers based NLP, LangChain for building chatbot flows, FAISS for similarity search, OCRs such as Tesseract OCR and TrOCR, and OpenCV and PIL for image processing.

## Chapter 2: Literature Review

### 2.1 Introduction

The implementation of AI in the healthcare industry has been widely investigated in recent years, mainly for intelligent machines which have been designed to help patients and professionals in the healthcare field alike. Some of these include the use of chatbots driven by AI due to their speed, availability, and affordability. This chapter discusses current studies on medical chatbots, NLP, multimodal systems, and document analysis.

### 2.2 AI-Powered Chatbots for Medicine

AI-powered chatbots employed in the domain of medicine replicate communication between patients and specialists working within the sphere. The initial designs of such software were largely rule-based. While simple in usage, these models had certain limitations in terms of being able to generate responses to more sophisticated queries.

Recent advancements in the development of applications have focused on AI-powered chatbots applying Machine Learning and Deep Learning techniques. Such apps can perceive user intent, comprehend language, and respond accordingly to the situation. Literature reports that these models may be beneficial for offering assistance with symptom assessment, mental disorders, and other medical services. However, the effectiveness of such applications depends upon their design.

### 2.3 Natural Language Processing in Healthcare

The ability to understand human language requires Natural Language Processing (NLP). Within the healthcare industry, NLP models are deployed in the extraction of vital information from patient queries, medical records, and medical literature.

Transformer-based models of NLP are currently at the forefront in the domain of language understanding. Such models can understand how words are used in their contexts; hence, they are capable of handling complex languages with medical terminology. However, there remain several challenges in natural language processing.

### 2.4 Retrieval-Augmented Generation (RAG) Models

Retrieval-Augmented Generation (RAG) is a relatively new technology combining the text retrieval and generation procedures into one. Unlike other approaches to generating text, RAG models do not rely solely on previously obtained knowledge but get their information from retrieval processes. In a medical context, this approach to building text-generating models is quite useful since it eliminates any chances of creating incorrect statements. Using vector embeddings to retrieve information via similarity searches, RAG models function perfectly for developing chatbots in the medical industry.

### 2.5 Optical Character Recognition in Healthcare

The Optical Character Recognition (OCR) technique is commonly applied in identifying texts which can be either scanned or handwritten. The healthcare sector utilizes OCR technology extensively for automated prescription, lab test results, and many other health documents.

Although OCR technology tends to function effectively on scanned text, it faces difficulties while working on handwritten text. However, with recent developments in OCR technology like the transformer OCR technique, it has become easier for OCR technology to handle handwritten text. OCR technology integration within the medical chatbot will help individuals submit their report files.

### 2.6 Multimodal AI Systems

Multimodal AI systems have been developed with a view to analyze and integrate various forms of information such as texts, images, documents, etc. It has been observed that the health care industry requires the use of both medical images and texts.

As per studies done in this regard, it has been found that using combinations of various forms of information leads to improved decision-making processes. However, the development of multimodal AI remains difficult owing to the requirement to integrate various models.

### 2.7 Limitations of Existing Chatbot Systems

Despite all the advancements achieved, existing chatbot systems for medical purposes continue to exhibit certain drawbacks. Some of them are listed below:

- The dependence of most chatbots on predefined data with poor adaptability

- The incapability to work with multimodal data like photos and handwritten notes
- Risk of providing users with incorrect or context-independent information
- Lack of personalization and inability to keep up with context while communicating

These are some of the disadvantages of existing medical chatbots.

2.8 Conclusion

Based on the analysis presented in this review, it becomes apparent that the application of AI-based medical chatbots will have a substantial effect on the efficiency of healthcare services delivery. The technologies such as NLP, RAG, OCR, and multimodal AI can help improve the functionalities of the medical chatbots proposed by researchers. However, current medical chatbots fail to demonstrate accuracy, flexibility, and versatility of inputs.

This paper suggests using various technologies to create a medical chatbot called ArogyaX.

## Chapter 3: Methodology

### 3.1 Medical AI Chatbot (RAG System)

The primary objective of this section is to develop a highly intelligent chatbot capable of answering medical queries. Rather than relying on mere speculation or outdated data by the AI, we designed the "Retrieval-Augmented Generation (RAG)" model, whereby the AI will search for real data from 15 medical books before responding.

3.1.1 Document Processing and Chunking

Before the AI can read the books, the raw PDFs must be processed and broken down.

- Cleaning: We remove page headers, footers, and non-text elements (like image captions) that don't add medical value.
- Chunking Logic: This one book alone exceeds the capacity of the AI's "short-term memory." This means first we'll break the text into chunks.
- Concept: We use a RecursiveCharacterTextSplitter. Let T be the total text. We create chunks $c_i$ with a size of 500 characters and an overlap of 50 characters.
- $$|c_i| = 500, \quad \text{Overlap} = |c_i \cap c_{i+1}| = 50$$

The overlap ensures that if a sentence like "Diabetes is caused by..." starts in one chunk, the conclusion isn't lost in the next.

3.1.2 Vectorization and Embedding

Computers cannot understand words; they only understand numbers. Vectorization is the process of turning a text chunk into a list of numbers (a vector) that represents its meaning.

- The Model: We use all-MiniLM-L6-v2, which transforms a text chunk into a 384-dimensional vector.
- The Transformation: If you provide a simple text S, the embedding function E maps it to a point in space.
- $$V = E(S) \in \mathbb{R}^{384}$$
- Semantic Mapping: In this 384-dimensional space, the vector for "Heart" is mathematically closer to "Cardiology" than it is to "Apple." This allows the system to understand relationships between medical terms.

### 3.1.3 Vector Database and JSON Export

Now, once we've collected all those thousands of vectors, we will need to come up with a method to store them.

- Storage: We utilize the vector database ChromaDB. For portability, we export the data in JSON format.

JSON Format:

```
{
 "id": "chunk_101",
 "text": "The left ventricle pumps blood to the aorta...",
 "metadata": {"book": "Gray's Anatomy", "page": 450},
 "vector": [0.124, -0.056, ..., 0.891]
}
```

### 3.1.4 Retrieval (Similarity Search)

- The process starts by converting the question asked by the user into a vector denoted as Vq, and then we find the most similar chunks within the database.
- Cosine Similarity: It is the angle between the question and the book.

The system generates the output based on top k chunks with the highest value.

## 3.2 Heart Rate Monitor (rPPG Model)

The rPPG model measures heart rate by tracking tiny color changes in the skin that the human eye cannot see.

### 3.2.1 Signal Extraction and Green Channel Analysis

We use a webcam to detect the face and isolate the forehead as our Region of Interest (ROI).

- Why Green? We isolate the Green (G) color channel because blood (hemoglobin) absorbs green light most strongly.
- Spatial Averaging: For every frame, we calculate the average brightness of the pixels in the forehead area to create a single data point g(t).

### 3.2.2 Signal Processing (Butterworth Filter)

The raw signal g(t) is "noisy" because of room lights or slight head movements. We apply a 2nd Order Butterworth Bandpass Filter.

- Frequency Range: We have set the limits between 0.7 Hz (42 BPM) and 3.0 Hz (180 BPM).
- Why we have used it? Because this filter will block other frequencies except for those within this range and provides the signal of "pulse" free of noise.

### 3.2.3 Fast Fourier Transform (FFT) and BPM

In order to calculate the heart rate, the FFT is used to transition from the "Time" domain to the "Frequency" domain.

- Heart Rate BPM: The system detects the peak frequency f_peak of the output data.
- Smoothing: We store the last 10 measurements and apply the Rolling Average calculation for smooth and human-understandable results to display on the screen.

## 3.3 Image Based Model

### 3.3.1 Data Collection Methods

The current proposal does not use any existing databases; rather, it uses a real-time multi-modal approach wherein user inputs are processed real-time.

Data sources utilized by the project include:

- User-Provided Inputs
    - One of the primary data sources used by the program involves obtaining data real-time through the user.
    - The possible user inputs include the following:

        - Text-based query for symptoms and medical condition
        - PDF medical reports
        - Medical imaging (X-ray, scanning, etc.)
- Synthetic Medical Reports
    - Some sample medical reports have been created artificially to assess the performance of the program.
    - Some of the simulated medical scenarios include the following:
        - Patient's demographic details
        - Diagnosis
        - Lab result
- Handwritten and Scanned Reports
    - The program can accept handwritten and scanned reports as input.
    - The program employs optical character recognition (OCR) technology.
- Image
    - Some image inputs, such as bone x-rays and kidney scans, have been manually provided to the system.
- The system is input-driven rather than dataset-driven, meaning it processes and analyzes data only when provided by the user.

### 3.3.2 Technologies and Tools Employed

The project leverages a variety of technologies that include NLP-related tools, computer vision techniques, and information retrieval technologies.

**Programming Language**

- Created using the programming language Python

**Core Technologies**

- **LangChain**
    - Used in building chatbot architecture
    - Creates conversation flow and context
- **Transformers (Hugging Face)**
    - Used in generating responses and comprehending questions about medicines
- **PyTorch**
    - Deep learning-based framework

**Document Processing**

- **PyMuPDF (pdfplumber)**
    - Extracts textual content from medical reports in PDF format
- **Tesseract OCR**
    - Extracts printed text from scanned documents
- **TrOCR**
    - Extracts handwritten text from medical reports

**Image Processing**

- **OpenCV**
    - Medical image pre-processing tool
- **PIL (Python Imaging Library)**
    - Image loading and formatting tool

**Information Retrieval and Storage**

- **FAISS (Facebook AI Similarity Search)**
    - Fast information retrieval mechanism based on vector storage using FAISS
- **Embedding Model**
    - Converts text into vectors

### 3.3.3 Data Description (System Input Description)

Since we have no particular dataset, inputs will be multimodal in nature. Following are the details related to system inputs.

**1. Text Input**

- The user enters the query based on their health-related problems through natural language.
- For example:
    - What is meant by high creatinine?

**2. PDF Reports**

- The user uploads real-time PDF reports during the use of an application
- Reports can be:
    - Results of blood tests
    - Medical diagnosis
- Handwritten and typed reports both

**3. Handwritten Reports**

- Includes prescriptions and handwritten notes provided by doctors
- Handled by TrOCR

**4. Medical Images**

- Can be:
    - Bone X-Ray Image
    - Kidney Image

**5. Data Preprocessing**

The following preprocessing is applied to input data from all types:

- Text preprocessing (noise removal)
- OCR processing
- Resizing images
- Data into embeddings

**3.3.4 Process Flow Diagram**

The process flow diagram is as follows:

1. **User Input**
    - Input provided in text/PDF/Image form
2. **Data Extraction**
    - OCR process carried out on input scanning/writing
    - Extracting text from PDFs
3. **Preprocessing**
    - Data preprocessing is done
4. **Embedding Generation**
    - Text embedding is carried out
5. **Information Search**
    - Search carried out using FAISS
6. **Response Generation**
    - Responses generated by NLP models
7. **Output Display**
    - Output display to the user

3.4 (General Medical RAG Engine)

3.4.1 Data Description (System Input & Knowledge Base)

While the report module focuses on real-time extraction, the general chatbot relies on a curated offline knowledge base paired with natural language user inputs.

User Text Input: The user enters health-related queries, symptoms, or casual greetings through natural language (e.g., "What are the symptoms of Type 2 Diabetes?").

Curated Knowledge Base: The system uses a pre-compiled, static vector database (chroma_db).

Dataset Composition: The database contains embedded chunks from 15 verified medical textbooks, ensuring all AI responses are grounded in clinical literature rather than internet scraping.

3.4.2 Technologies & Tools Employed

General purpose chat is implemented based on the latest advancements in Information Retrieval Technologies and Large Language Models, mimicking the work of medical librarian chatbot.

Programming Language: Python

Critical Technologies:

LangChain: For developing RAG application, creating prompt templates and managing conversation memory buffer.

Groq API: Brings intelligence powered by the llama-3.3-70b-versatile model.

Transformers (HuggingFace): Implements all-MiniLM-L6-v2 model for mapping of the query to its dense vectors representation.

Information Retrieval & Storage:

ChromaDB: Vector database to store medical textbook embeddings.

Backend Development Framework:

FastAPI: Web framework which acts as an intermediary between Python AI backend and frontend.

3.3 Process Flow Diagram

The process flow for generic architecture of RAG will be as follows:

User Input

Input provided by users is in text format through a chat interface.

Intent Classification & Query Rewriting

The system first captures the raw input from the user and processes it through the “Rewriter” LLM chain.

Semantic Routing: The chain automatically classifies the user's intent. If the system finds out it's chitchat ("Hello"), it routes the user to a light chitchat pipeline.

Search Optimization: If the system recognizes that there's a medical question asked, it converts the chitchat into an optimized query to the database.

Embedding Generation

The best search query is represented as a vector embedding with Hugging Face.

Information Retrieval using Maximum Marginal Relevance (MMR)

Carried out using ChromaDB as the tool for information retrieval.

Maximum Marginal Relevance (MMR): Instead of getting only the 5 closest texts (since this approach will cause repetitions), the procedure involves using the MMR approach where first, 20 texts are retrieved as candidate texts, but then only 5 texts that are relevant and yet diversified mathematically so that the LLM receives a comprehensive view of the subject.

Context Construction

Textbook texts are "upgraded" together with the query and recent chat session in a strict system prompt.

Response Generation

Llama 3 model is responsible for responding. In terms of prompts, there is a strict requirement for LLMs to generate answers strictly based on the textbook context without any hallucinations whatsoever.

Output Display

The output is shown to the user in a formatted form using Markdown, which contains the Title and Page Number of the Textbook utilized to generate the answer.

## Chapter 4: Outcomes, Results and Discussion

## Dataset used for Chunking And Chunking Results

```
BOOKS:
  Total books: 11
   1. Netter's Atlas of Human Anatomy 7th Ed.
   2. Braunwald's Heart Disease
   3. Davidson's Principles and Practice of Medicine
   4. Goodman & Gilman's Pharmacology
   5. Gray's Anatomy for Students 3rd Ed.
   6. Guyton Textbook of Medical Physiology
   7. Harrison's Principles of Internal Medicine 21st Ed.
   8. General Medical Book
   9. Nelson Textbook of Pediatrics
  10. Oxford Handbook of Oncology
  11. Williams Obstetrics 24th Ed.

CHUNKS:
  Total chunks: 149,398
  Total characters: 65,880,275
  Average chunk size: 441 chars

VECTOR STORE:
  Collection: medical_books_15
  Total vectors: 149,398
  Embedding model: sentence-transformers/all-MiniLM-L6-v2
  Persist directory: /content/drive/MyDrive/medical_chroma_db
```

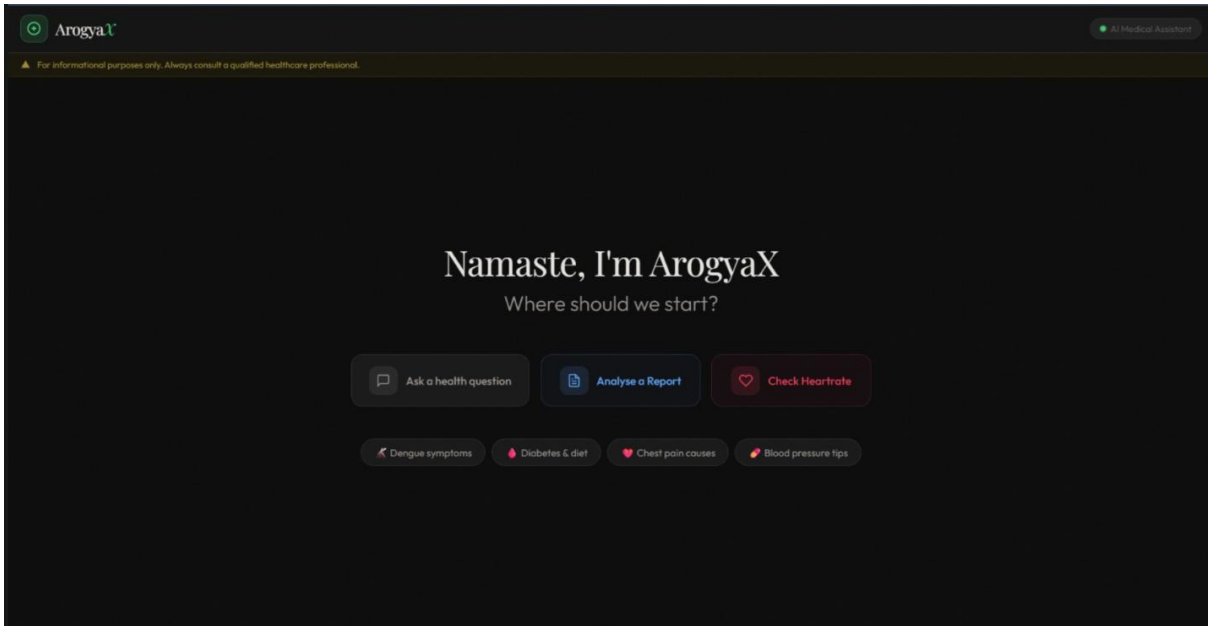
```
CHUNKS:
  Total chunks: 149,398
  Total characters: 65,880,275
  Average chunk size: 441 chars

VECTOR STORE:
  Collection: medical_books_15
  Total vectors: 149,398
  Embedding model: sentence-transformers/all-MiniLM-L6-v2
  Persist directory: /content/drive/MyDrive/medical_chroma_db

CHUNKS PER BOOK:
  Nelson Textbook of Pediatrics                    51,823 (34.69%)
  Braunwald's Heart Disease                        31,492 (21.08%)
  Goodman & Gilman's Pharmacology                  19,619 (13.13%)
  Williams Obstetrics 24th Ed.                     16,499 (11.04%)
  Guyton Textbook of Medical Physiology            11,286 ( 7.55%)
  Gray's Anatomy for Students 3rd Ed.               7,330 ( 4.91%)
  General Medical Book                              5,961 ( 3.99%)
  Netter's Atlas of Human Anatomy 7th Ed.           2,788 ( 1.87%)
  Oxford Handbook of Oncology                       2,538 ( 1.70%)
  Davidson's Principles and Practice of Medicine       62 ( 0.04%)

FILES:
  JSON export: 129.06 MB
  ChromaDB: 994.29 MB
======================================================================
```

4.1 Queries And Their Outputs



ArogyaX

AI Medical Assistant

For informational purposes only. Always consult a qualified healthcare professional.

Hello! There.
Feel free to ask me about any disease, symptoms, or cure.

hey can you tell me about types of skin diseases

**Types of Skin Diseases**

There are various types of skin diseases, including:

- **Cutaneous Fungal Infections**: such as kerion, tinea capitis, and dermatophyte diseases
- **Eczematous Dermatitis:** associated with **Atopic Dermatitis (AD)**, characterized by pruritus, eczematous dermatitis, and a chronic or chronically relapsing course
- **Mastocytoses:** a group of disorders characterized by aggregates of mast cells in the dermis, including:
  - Solitary mastocytoma
  - Urticaria pigmentosa (2 forms: childhood variant and adult-onset)
  - Diffuse cutaneous mastocytosis
  - Telangiectasia macularis eruptiva perstans

Ask a health question...

ArogyaX can make mistakes. Verify with a doctor.

ArogyaX

AI Medical Assistant

For informational purposes only. Always consult a qualified healthcare professional.

i have rashes in my hand, deep dark red colour, what can be the reason?

**Possible Causes of Dark Red Rashes on Hands**

- **Viral infections**: Certain viral infections can cause exanthematous eruptions, which may appear as dark red rashes on the hands.
- **Bacterial cellulitis**: This is a bacterial infection of the skin and underlying tissues, which can cause redness, swelling, and warmth on the affected area, including the hands.
- **Petechiae and purpura**: These are small, dark red or purple spots on the skin that can occur due to low platelet counts or damage to small blood vessels.

**Specific Conditions**

- **Hand-foot-and-mouth disease**: This is a viral infection that can cause a typical rash on the hands and feet, although it is usually more common in children.
- **Equine antithymocyte globulin reaction**: This is a rare condition where an initial rash can occur on the hands and feet, starting as a thin serpiginous band of erythema, which can be replaced by petechiae or purpura.

**Medication-Related Rashes**

- **Erythromycin topical**: This medication can cause burning, dry skin, hives, red and peeling skin, which may appear as dark red rashes on the hands.

Ask a health question...

ArogyaX

Report Mode

AI Medical Assistant

For informational purposes only. Always consult a qualified healthcare professional.

**Report loaded:** WhatsApp Image 2026-04-20 at 7.16.55 PM.jpeg

You can now ask me to **summarize this report**, explain findings, or ask any specific question. I'll also draw from my medical textbook knowledge.

what is the report about?

The patient's report appears to be a radiological findings report, specifically an X-ray report, that lists several abnormalities in the patient's lungs and chest cavity. The findings include:

1. Atelectasis (collapse or closure of a lung resulting in reduced or absent gas exchange)
2. Infiltration (presence of abnormal substances or cells in the lung tissue)
3. Mass (a lump or growth in the lung)
4. Pneumothorax (air or gas in the pleural cavity, which can cause the lung to collapse)
5. Consolidation (an area of lung tissue that is filled with something other than air, such as pus, fluid, or other material)
6. Emphysema (a condition in which the air sacs in the lungs are damaged, leading to breathlessness)
7. Fibrosis (scarring of lung tissue)

WhatsApp Image 2026-04-20 at 7.16.55 PM.jpeg

Ask about this report...

ArogyaX can make mistakes. Verify with a doctor.

what is this image about

Based on the patient's report findings, the image appears to be a scan result that shows "No Tumor". This suggests that the patient has undergone a medical imaging procedure, such as an MRI or CT scan, and the results indicate that no tumor was detected.

As a medical AI, I can tell you that this is a positive finding, as it rules out the presence of a tumor. However, it's essential to consider this result in the context of the patient's overall medical history and symptoms.

Please note that I am an AI, not a doctor, and this interpretation should not be considered a substitute for a professional medical diagnosis or advice. A qualified healthcare professional should be consulted to discuss the patient's specific condition and any further necessary steps.

**Sources:**

Uploaded Patient Report

### 4.2 Result Analysis

The performance of the chatbot in terms of comprehending users' questions and providing appropriate responses was effective.

It was found out that the system functions effectively for medical questions and also for queries based on symptoms.

Since the use of natural language processing techniques is incorporated in the design of the bot,

the system can understand natural language questions very effectively.

But response will differ depending upon the question asked by the user.

### 4.3 Observations

If the queries are clear and specific, the responses provided by the chatbot will be more accurate

General questions relating to medicine are better handled than complex ones

Response times are fast if the input provided is written

There will be consistency in the operation of the system regardless of how many tests are conducted

### 4.4 Verification of Results

To confirm the results, the system was tested using different questions from users.

Responses provided by the chatbot were compared against the correct medical data.

It was discovered that the response provided was always relevant and appropriate.

The system was tested under different conditions.

### 4.5 Key Findings and Insights

The chatbot effectively shows the use of AI in assisting with health-related needs.

The chatbot can provide fast and easy access to medical information.

Using multimodal input increases the usability of the system.

It can serve as a first point of assistance for the user.

However, it does not replace the need for expert medical advice.

# Chapter 5: Conclusion and Future Work Plan

## 5.1 Conclusion

It is apparent from the description of the implementation of the ArogyaX – AI Medical Chatbot that it is possible to effectively use Artificial Intelligence technologies in the field of medicine in order to facilitate the process of obtaining healthcare services. Due to the integration of NLP, OCR, and information retrieval approaches, the program will be capable of understanding user requests and giving adequate responses.

The software is designed to work with text requests, medical records, and photos submitted by users. The use of a retrieval-based approach allows improving the effectiveness of responses due to the inclusion of context in their generation. The performance of the program was examined using a set of various test cases and turned out to be effective in producing health-related answers.

In practice, this innovative technology creates a great possibility for increasing the availability of healthcare services and avoiding unnecessary visits to hospitals because of trivial medical conditions. Still, the software created cannot be used as a replacement for the advice of medical specialists.

## 5.2 Future Work Plan

Even though the chatbot is working well, there are certain aspects that need to be improved in the future:

Improving the accuracy of responses by including more sophisticated and subject-specific medical models

Improving OCR capabilities, particularly the ability to recognize complicated handwritten medical prescriptions

Improving the capability of the chatbot to analyze various medical images in detail

Including a voice-based feature that will allow users to access the chatbot without having to type messages

Incorporation of live medical databases or APIs for obtaining updated data

Inclusion of user history and personalization for generating accurate responses

Deployment of the chatbot as an app on users' smartphones

## 6. References

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